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Joint connector

Abstract

There is provided a joint connector including: a ground terminal member integrally formed by a plate-shaped fixing portion fixed to a ground portion, a body horizontal plate portion extending from the plate-shaped fixing portion, a body rising plate portion formed by bending in a rising direction, a terminal connection portion provided formed with a plurality of terminal connecting portions electrically connectable with a counterpart terminal, and a coupling piece; and a housing integrally formed by a housing body portion in which the body rising plate portion and the terminal connection portion are insert-molded in a state where the plurality of terminal connecting portions are protruded for connecting the counterpart terminal, and a reinforcing rib portion provided to couple a resin portion covering the coupling piece and a resin portion covering the terminal connection portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2022-130186 filed on Aug. 17, 2022, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a joint connector.

BACKGROUND ART

(3) A known joint connector for collectively connecting a plurality of electric wires to a ground portion of a vehicle includes: a conductive plate (ground terminal member) having a body side grounding portion (plate-shaped fixing portion) which is fixed to a body (ground portion) of the vehicle by screws, and a connecting portion at one end of the body side grounding portion in which a plurality of terminal connecting portions to which wire-side terminals (counterpart terminals) are electrically connected are formed; and a resin housing in which the conductive plate is insert-molded (for example, see JP2013-065502 A).

(4) The conductive plate is formed by integrally forming the body side grounding portion and the connecting portion by pressing a conductive metal plate. A bolt hole through which a ground bolt can be inserted is formed in the body side grounding portion. The connecting portion has a connection portion that forms a substantially horizontal U-shape, and the plurality of terminal connecting portions are formed to protrude from a side edge of the connection portion. The connection portion is coupled to one end of the body side grounding portion via a step portion (bent portion) rising upward from the side edge of the body side grounding portion.

(5) Finally, the connection portion and a part of the step portion of the conductive plate are insert-molded in the housing.

(6) However, in the joint connector described above, vibration is applied to the joint connector due to vibration during traveling and the like of the vehicle, and the joint connector swings around the step portion, which is the bent portion of the conductive plate. Then, elastic deformation repeatedly occurs at the step portion (bent portion) of the conductive plate, and deformation and breakage may occur in the step portion.

SUMMARY OF INVENTION

(7) The present disclosure provides a joint connector capable of preventing deformation and breakage occurring in a bent portion of a ground terminal member.

(8) According to an illustrative aspect of the present disclosure, a joint connector includes: a ground terminal member made of a conductive metal; and a housing made of an insulating resin. The ground terminal member includes: a plate-shaped fixing portion that is fixed to a ground portion of a vehicle; a body horizontal plate portion that extends from the plate-shaped fixing portion; a body rising plate portion that is formed by bending the body horizontal plate portion in a rising direction at a bent portion; a terminal connection portion that is provided above the body rising plate portion and that is formed with a plurality of terminal connecting portions electrically connectable with a counterpart terminal; and a coupling piece that is continuously provided with the body horizontal plate portion. The ground terminal member is configured by integrally forming the plate-shaped fixing portion, the body horizontal plate portion, the body rising plate portion, the terminal connection portion, and the coupling piece. The housing includes: a housing body portion in which the body rising plate portion and the terminal connection portion are insert-molded in a state in which the plurality of terminal connecting portions are protruded for connecting the counterpart terminal; and a reinforcing rib portion provided to couple a first resin portion covering the coupling piece and a second resin portion covering the terminal connection portion. The housing is configured by integrally forming the housing body portion and the reinforcing rib portion.

(9) The present disclosure is briefly described above. Further, details of the present disclosure will be further clarified by reading through a mode for carrying out the disclosure described below (hereinafter referred to as an “embodiment”) with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view of a joint connector according to an embodiment of the present disclosure to which a counterpart connector is fitted;

- (2) FIG. 2 is an exploded perspective view of the joint connector and the counterpart connector shown in FIG. 1;
- (3) FIG. 3 is a perspective view of the joint connector shown in FIG. 2 as viewed from a front side;
- (4) FIG. 4 is a perspective view of the joint connector in which a ground terminal member is insert-molded in a housing;
- (5) FIG. 5 is a perspective view of the ground terminal member as viewed from a rear upper side;
- (6) FIG. 6 is a perspective view of the ground terminal member as viewed from a front lower side;
- and
- (7) FIG. 7 is a longitudinal sectional view of the counterpart connector and the joint connector which is screwed and fixed to a ground portion of a vehicle.

DESCRIPTION OF EMBODIMENTS

- (8) A specific embodiment of the present disclosure will be described below with reference to the drawings.
- (9) FIG. 1 is a perspective view of a joint connector **1** according to an embodiment of the present disclosure to which a counterpart connector **51** is fitted. FIG. 2 is an exploded perspective view of the joint connector **1** and the counterpart connector **51** shown in FIG. 1.
- (10) Note that, in the present description, a front-rear direction, an upper-lower direction, and a left-right direction of the joint connector **1** follow directions of arrows shown in FIG. 1. That is, the front-rear direction is a direction along a connector fitting direction of the joint connector **1**, and a side to which a counterpart housing **52** is fitted is defined as a front side of a housing **31**. Further, the upper-lower direction is a short direction of the housing **31** orthogonal to the connector fitting direction of the flat rectangular parallelepiped joint connector **1**, and a lock portion **39** side of the housing **31** is defined as an upper side. Further, the left-right direction is a longitudinal direction of the housing **31** orthogonal to the connector fitting direction of the joint connector **1**.
- (11) As shown in FIGS. 1 and 2, the counterpart connector **51** is fitted to the joint connector **1** according to the present embodiment.
- (12) The joint connector **1** includes a ground terminal member **11** made of conductive metal and the housing **31** made of an insulating resin. The counterpart connector **51** includes terminal-attached electric wires **61**, a counterpart housing **52** made of an insulating resin, and an annular packing **41**.
- (13) A fitting recessed portion **33** is formed in the housing **31** so that a front end side is open, which is the front side in the fitting direction with the counterpart connector **51** (see FIG. 3). As shown in FIG. 2, a counterpart fitting portion **53** is formed on a front end side of the counterpart housing **52** to be fitted to the joint connector **1**. The counterpart connector **51** is fitted to the joint connector **1** by fitting the counterpart fitting portion **53** of the counterpart housing **52** into the fitting recessed portion **33** of the housing **31**.
- (14) As shown in FIG. 2, a plurality of terminal accommodating chambers **55** are formed in the counterpart fitting portion **53** of the counterpart housing **52**. The terminal accommodating chambers **55** are formed along the fitting direction with the joint connector **1**. The terminal accommodating chambers **55** are arranged in a left-right width direction of the counterpart housing **52**, and are arranged in two stages in the upper-lower direction. The counterpart fitting portion **53** has lance portions **56** protruding into respective terminal accommodating chambers **55** (see FIG. 7). A lock arm **57** is formed on an upper face of the counterpart housing **52**.
- (15) The counterpart housing **52** has a hood portion **54** around the counterpart fitting portion **53**, and an annular packing **41** is fitted into the hood portion **54**. When the counterpart fitting portion **53** of the counterpart housing **52** is fitted into the fitting recessed portion **33** of the housing **31**, the annular packing **41** stops water not to enter a part between the fitting recessed portion **33** and the counterpart fitting portion **53**.
- (16) A counterpart terminal **63** of the terminal-attached electric wire **61** is accommodated in the terminal accommodating chamber **55** from a rear side of the counterpart housing **52**. The counterpart terminal **63** is formed of, for example, a conductive metal material such as copper or a

copper alloy, and is crimped and conductively connected to a terminal of an electric wire **62** in which periphery of a conductor is covered with an outer sheath.

(17) The counterpart terminal **63** is locked to the lance portion **56** by being inserted into the terminal accommodating chamber **55** from the rear side of the counterpart housing **52**. As a result, the counterpart terminal **63** is held in a state of being accommodated in the terminal accommodating chamber **55**.

(18) A waterproof plug **64**, which is a cylindrical packing, is fitted to a terminal of the electric wire **62**. When the terminal of the electric wire **62** is inserted into the terminal accommodating chamber **55**, the waterproof plug **64** comes into close contact with an inner peripheral face of the terminal accommodating chamber **55** to waterproof a space between inner peripheral face and the electric wire **62** in a watertight manner.

(19) FIG. **3** is a perspective view of the joint connector shown **1** in FIG. **2** as viewed from the front side. FIG. **4** is a perspective view of the joint connector **1** in which the ground terminal member **11** is insert-molded in the housing **31**. FIG. **5** is a perspective view of the ground terminal member **11** as viewed from a rear upper side. FIG. **6** is a perspective view of the ground terminal member **11** as viewed from a front lower side.

(20) As shown in FIGS. **3** and **4**, in the joint connector **1**, the ground terminal member **11** is insert-molded in the housing **31**.

(21) As shown in FIGS. **5** and **6**, the ground terminal member **11** is integrally formed with a plate-shaped fixing portion **12**, a body horizontal plate portion **15** extending from the plate-shaped fixing portion **12**, a body rising plate portion **19** extending from the body horizontal plate portion **15**, a terminal connection portion **21** provided above the body rising plate portion **19**, a crank-shaped bent portion **28** formed by bending the body rising plate portion **19** in the vicinity of the terminal connection portion **21** in a crank shape, and coupling pieces **16** continuously provided on left and right side edges of the body horizontal plate portion **15** and protruding upward.

(22) A bolt through hole **13** is formed in the plate-shaped fixing portion **12**. The plate-shaped fixing portion **12** is screwed and fixed to a body **100**, which is a ground portion of the vehicle, by fastening a stud bolt **71** inserted into the bolt through hole **13** to the body **100** with a nut **73**. Thus, the ground terminal member **11** is grounded to the body **100**. A rotation prevention piece **14** is formed in a part of the plate-shaped fixing portion **12**. The rotation prevention piece **14** restricts, by engaging with a step or a hole portion around a ground surface to which the plate-shaped fixing portion **12** is screwed and fixed, rotation of the plate-shaped fixing portion **12** with respect to the ground surface.

(23) A pair of coupling pieces **16** bent upward are continuously provided on the left and right edges of the body horizontal plate portion **15** extending forward from the plate-shaped fixing portion **12**. The coupling pieces **16** are each formed with a through hole **18**, which is a locking portion for increasing a coupling strength between the coupling pieces **16** and resin portions **38** of the housing **31** covering the coupling pieces **16**.

(24) The body rising plate portion **19** is formed by bending the body horizontal plate portion **15** at 90° in a rising direction at a bent portion **20**. A bending angle of the bent portion **20** is not limited to 90°, and can be appropriately changed according to a shape around the ground surface to which the plate-shaped fixing portion **12** is screwed and fixed, and the fitting direction of the counterpart connector **51**.

(25) Further, the bent portion **20** is provided with an embossed portion **27** extending over the body horizontal plate portion **15** and the body rising plate portion **19**.

(26) At both left and right end portions of the terminal connection portion **21**, folding plate portions **22** are continuously provided to the terminal connection portion **21** respectively, and the folding plate portion **22** folded forward is overlapped with the terminal connection portion **21**.

(27) Dowels **25** protruding toward the terminal connection portion **21** side are formed in the respective folding plate portions **22** by dowel processing. By fitting the dowels **25** into holes **24**

formed in the terminal connection portion **21**, the folding plate portions **22** are restricted from being displaced with respect to the terminal connection portion **21**.

(28) A plurality of terminal connecting portions **26** protrude in the same direction from upper and lower edges of the folding plate portion **22**. In the ground terminal member **11**, the plurality of terminal connecting portions **26** protruding forward in a plane perpendicular to a plane of the body rising plate portion **19** are arranged in two stages in the upper-lower direction.

(29) As shown in FIGS. **3** and **4**, in the housing **31** in which the ground terminal member **11** is insert-molded, a housing body portion **32** including the lock portion **39** and the fitting recessed portion **33** into which the counterpart fitting portion **53** of the counterpart housing **52** is fitted, and a pair of reinforcing rib portions **35** are integrally molded.

(30) That is, in the housing body portion **32**, the body rising plate portion **19** and the terminal connection portion **21** are insert-molded in a state in which the plurality of terminal connecting portions **26** protrude into the fitting recessed portion **33**. Further, the pair of reinforcing rib portions **35** are integrally formed with the housing body portion **32** so as to connect the resin portions **38** covering the coupling pieces **16** and a resin portion **36** covering the terminal connection portion **21**. The resin portions **38** covering the coupling pieces **16** flow into the through holes **18** of the coupling pieces **16** and solidifies, thereby increasing coupling strength between the coupling pieces **16** and the resin portions **38**.

(31) Next, a case where the counterpart connector **51** is fitted to the joint connector **1** will be described.

(32) FIG. **7** is a longitudinal sectional view of the counterpart connector **51** and the joint connector **1** which is screwed and fixed to the body **100** of the vehicle.

(33) In order to fit the counterpart connector **51** to the joint connector **1** which is screwed and fixed to the body **100** of the vehicle, a front end of the counterpart housing **52** of the counterpart connector **51** is brought close to a front end of the housing **31** of the joint connector **1**. Then, the counterpart fitting portion **53** of the counterpart housing **52** is inserted into the fitting recessed portion **33** of the housing **31**.

(34) Then, as shown in FIG. **7**, the terminal connecting portions **26** formed on the ground terminal member **11** of the joint connector **1** are conductively connected to the counterpart terminals **63** accommodated in the terminal accommodating chamber **55** of the counterpart housing **52**, and the electric wires **62** are electrically connected to the ground terminal member **11**. The electric wires **62** are also electrically connected to each other via the ground terminal member **11**. In this state, the lock arm **57** locks the lock portion **39**, and the housing **31** and the counterpart housing **52** are locked to each other in a state of being fitted.

(35) Therefore, according to the joint connector **1** of the present embodiment described above, the coupling pieces **16** and the terminal connection portion **21** of the ground terminal member **11** are insert-molded in the housing **31**, so that the resin portions **38** covering the coupling pieces **16** and the resin portion **36** covering the terminal connection portion **21** are coupled by the reinforcing rib portions **35**. That is, the body horizontal plate portion **15** and the body rising plate portion **19** are obliquely reinforced by the reinforcing rib portions **35** like an angle brace.

(36) Therefore, even when the joint connector **1** tends to swing about the bent portion **20** of the ground terminal member **11** as a fulcrum, an angle formed by the body horizontal plate portion **15** and the body rising plate portion **19** is held by the reinforcing rib portions **35**, thereby restricting the swinging. Therefore, elastic deformation does not occur repeatedly at the bent portion **20** of the ground terminal member **11**.

(37) Accordingly, even if vibration is applied to the joint connector **1** due to vibration during traveling and the like of the vehicle, deformation and breakage occurring in the bent portion **20** of the ground terminal member **11** can be prevented.

(38) Further, according to the joint connector **1** of the present embodiment, the coupling strength between the coupling piece **16** and the resin portion **38** covering the connecting piece **16** is

increased by the through hole **18** provided in the coupling piece **16**. Therefore, it is possible to prevent the reinforcing rib portion **35** from coming off the coupling piece **16** due to vibration or the like applied to the joint connector **1**.

(39) The locking portion of the present disclosure is not limited to the through hole **18**, and various forms such as a notch and a bending protrusion can be adopted based on the gist of the present disclosure.

(40) Further, according to the joint connector **1** of the present embodiment, the bent portion **20** is provided with the embossed portion **27** extending over the body horizontal plate portion **15** and the body rising plate portion **19**. Therefore, when the joint connector **1** is handled by conveyance or the like, even if an external force is applied to the ground terminal member **11**, deformation of the bent portion **20** is prevented, and the angle formed by the body horizontal plate portion **15** and the body rising plate portion **19** is maintained.

(41) Further, according to the joint connector **1** of the present embodiment, the housing **31** includes, in the housing body portion **32**, the fitting recessed portion **33** into which the counterpart fitting portion **53** of the counterpart housing **52** having the plurality of terminal accommodating chambers **55** each accommodating the counterpart terminal **63** is fitted, and the plurality of terminal connecting portions **26** protrude into the fitting recessed portion **33**. Therefore, by screwing and fixing the joint connector **1** to the body **100** of the vehicle and fitting the counterpart fitting portion **53** of the counterpart housing **52** into the fitting recessed portion **33** of the housing body portion **32**, the plurality of electric wires **62** electrically connected to the counterpart terminals **63** can be collectively grounded to the body **100** of the vehicle via the plurality of terminal connecting portions **26** of the ground terminal member **11**.

(42) Therefore, according to the joint connector **1** of the present embodiment, it is possible to prevent deformation and breakage occurring in the bent portion **20** of the ground terminal member **11**.

(43) It should be noted that the present disclosure is not limited to the above embodiment, and modifications, improvements, and the like can be made as appropriate. In addition, materials, shapes, dimensions, numbers, arrangement locations, and the like of the respective constituent elements in the above embodiment are optionally selected and are not limited as long as the present disclosure can be implemented.

(44) The housing **31** of the above-described embodiment, includes, in the housing body portion **32**, the fitting recessed portion **33** into which the counterpart fitting portion **53** of the counterpart housing **52** having the plurality of terminal accommodating chambers **55** each accommodating the counterpart terminal **63** is fitted. The terminal connecting portions **26** protruding into the fitting recessed portion **33** are electrically connected to the counterpart terminals **63** of the counterpart housing **52**, but the present disclosure is not limited thereto. For example, the housing may include a terminal holding mechanism such as a locking lance or a retainer for holding the counterpart terminal **63** in the housing body portion, and the plurality of terminal connecting portions may be electrically connected to the counterpart terminal held by the terminal holding mechanism.

(45) According to a first aspect of the present disclosure, a joint connector (**1**) includes: a ground terminal member (**11**) made of a conductive metal; and a housing (**31**) made of an insulating resin. The ground terminal member (**11**) includes: a plate-shaped fixing portion (**12**) that is fixed to a ground portion of a vehicle (body **100**); a body horizontal plate portion (**15**) that extends from the plate-shaped fixing portion (**12**); a body rising plate portion (**19**) that is formed by bending the body horizontal plate portion (**15**) in a rising direction at a bent portion (**20**); a terminal connection portion (**21**) that is provided above the body rising plate portion (**19**) and that is formed with a plurality of terminal connecting portions (**26**) electrically connectable with a counterpart terminal (**63**); and a coupling piece (**16**) that is continuously provided with the body horizontal plate portion (**15**). The ground terminal member (**11**) is configured by integrally forming the plate-shaped fixing portion (**12**), the body horizontal plate portion (**15**), the body rising plate portion (**19**), the terminal

connection portion (21), and the coupling piece (16). The housing (31) includes: a housing body portion (32) in which the body rising plate portion (19) and the terminal connection portion (21) are insert-molded in a state in which the plurality of terminal connecting portions (26) are protruded for connecting the counterpart terminal (63); and a reinforcing rib portion (35) provided to couple a first resin portion (38) covering the coupling piece (16) and a second resin portion (36) covering the terminal connection portion (21). The housing (31) is configured by integrally forming the housing body portion (32) and the reinforcing rib portion (35).

(46) According to the joint connector (1) having the configuration of the first aspect, the coupling pieces (16) and the terminal connection portion (21) of the ground terminal member (11) are insert-molded in the housing (31), so that the resin portions (38) covering the coupling pieces (16) and the resin portion (36) covering the terminal connection portion (21) are coupled by the reinforcing rib portions (35). That is, the body horizontal plate portion (15) and the body rising plate portion (19) are obliquely reinforced by the reinforcing rib portions (35) like an angle brace.

(47) Therefore, even when the joint connector (1) tends to swing about the bent portion (20) of the ground terminal member (11) as a fulcrum, an angle formed by the body horizontal plate portion (15) and the body rising plate portion (19) is held by the reinforcing rib portions (35), thereby restricting the swinging. Therefore, elastic deformation does not occur repeatedly at the bent portion (20) of the ground terminal member (11).

(48) Accordingly, even if vibration is applied to the joint connector (1) due to vibration during traveling and the like of the vehicle, deformation and breakage occurring in the bent portion (20) of the ground terminal member (11) can be prevented.

(49) According to a second aspect of the present disclosure, the coupling piece (16) has a locking portion (through hole 18) for increasing a coupling strength between the coupling piece (16) and the first resin portion (38) covering the coupling piece (16).

(50) According to the joint connector (1) having the configuration of the second aspect, the coupling strength between the coupling piece (16) and the resin portion (38) covering the connecting piece (16) is increased by the locking portion (through hole 18) provided in the coupling piece (16). Therefore, it is possible to prevent the reinforcing rib portion (35) from coming off the coupling piece (16) due to vibration or the like applied to the joint connector (1).

(51) According to a third aspect of the present disclosure, the bent portion (20) is provided with an embossed portion (27) extending over the body horizontal plate portion (15) and the body rising plate portion (19).

(52) According to the joint connector (1) having the configuration of the third aspect, when the joint connector (1) is handled by conveyance or the like, even if an external force is applied to the ground terminal member (11), deformation of the bent portion (20) is prevented, and the angle formed by the body horizontal plate portion (15) and the body rising plate portion (19) is maintained.

(53) According to a fourth aspect of the present disclosure, the housing body portion (31) includes a fitting recessed portion (33) configured to fit with a counterpart fitting portion (53) of a counterpart housing (52) having a plurality of terminal accommodating chambers (55) each accommodating the counterpart terminal (63). The plurality of terminal connecting portions (26) protrude inside the fitting recessed portion (33).

(54) According to the joint connector (1) having the configuration of the fourth aspect, by screwing and fixing the joint connector (1) to the ground portion (body 100) of the vehicle and fitting the counterpart fitting portion (53) of the counterpart housing (52) into the fitting recessed portion (33) of the housing body portion (32), the plurality of electric wires (62) electrically connected to the counterpart terminals (63) can be collectively grounded to the ground portion (body 100) of the vehicle via the plurality of terminal connecting portions (26) of the ground terminal member (11).

(55) According to the joint connector of the present disclosure, it is possible to prevent deformation and breakage occurring in a bent portion of a ground terminal member.

Claims

1. A joint connector comprising: a ground terminal member made of a conductive metal; and a housing made of an insulating resin, wherein the ground terminal member includes: a plate-shaped fixing portion that is fixed to a ground portion of a vehicle; a body horizontal plate portion that extends from the plate-shaped fixing portion; a body rising plate portion that is formed by bending the body horizontal plate portion in a rising direction at a bent portion; a terminal connection portion that is provided above the body rising plate portion and that is formed with a plurality of terminal connecting portions electrically connectable with a counterpart terminal; and a coupling piece that is continuously provided with the body horizontal plate portion, the ground terminal member is configured by integrally forming the plate-shaped fixing portion, the body horizontal plate portion, the body rising plate portion, the terminal connection portion, and the coupling piece, the housing includes: a housing body portion in which the body rising plate portion and the terminal connection portion are insert-molded in a state in which the plurality of terminal connecting portions are protruded for connecting the counterpart terminal; and a reinforcing rib portion provided to couple a first resin portion covering the coupling piece and a second resin portion covering the terminal connection portion, and the housing is configured by integrally forming the housing body portion and the reinforcing rib portion.
 2. The joint connector according to claim 1, wherein the coupling piece has a locking portion for increasing a coupling strength between the coupling piece and the first resin portion covering the coupling piece.
 3. The joint connector according to claim 1, wherein the bent portion is provided with an embossed portion extending over the body horizontal plate portion and the body rising plate portion.
 4. The joint connector according to claim 1, wherein the housing body portion includes a fitting recessed portion configured to fit with a counterpart fitting portion of a counterpart housing having a plurality of terminal accommodating chambers each accommodating the counterpart terminal, and the plurality of terminal connecting portions protrude inside the fitting recessed portion.
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